

Assignment 4 – Reflection Essay on Transformer

Overview: In this assignment, you will read and reflect on the foundational paper "Attention Is All You Need" by Ashish Vaswani et al. (2017). This paper introduces the Transformer model, a key innovation in machine learning that powers many modern AI systems, such as language translation and chatbots. The focus is on building your understanding of ML concepts through critical reflection.

Learning Objectives:

- Grasp the main ideas behind the Transformer model.
- Think about its strengths, limitations, and real-world impact.
- Practice reading academic papers and expressing your own thoughts in writing.

Instructions:

- **Read the Paper:** Read the paper "Attention Is All You Need" (available on arXiv: <https://arxiv.org/abs/1706.03762>). Pay special attention to the introduction, model architecture (including self-attention and positional encoding), and key results. Don't worry if some math is challenging—focus on the big ideas. Take notes on what stands out to you.
- **Write a Reflection Essay:** Write a reflection essay of **no more than two pages** (double-spaced, approximately 500-750 words). This is **not a summary**; instead, share your thoughts and analysis. Structure it with an introduction, body, and conclusion. Use these prompts to guide your writing (cover them naturally, not as a list):
 - o What makes the Transformer different from older models, such as RNNs? Why might "attention is all you need" be a bold claim?
 - o Explain the self-attention mechanism in simple terms and why it's useful for tasks like processing sentences.
 - o How has the Transformer changed ML? Give examples like ChatGPT or image models that use similar ideas.
 - o Your personal take: What was surprising or confusing? What pros and cons do you see (e.g., speed vs. resource use)? Connect it to class topics if relevant.

Formatting and Submission

- Use a standard font (e.g., Times New Roman, 12 pt) with 1-inch margins.
- Include your name, student ID, and assignment title at the top.
- Cite the paper and any other sources in APA or MLA style. Include a short references section.

Evaluation Criteria:

- **Understanding of Key Concepts (40 points):** Shows a clear grasp of the paper's main ideas without just repeating them.

- **Reflection and Insights (30 points):** Includes your own opinions, connections, and balanced views on strengths/weaknesses.
- **Organization and Writing (20 points):** Clear structure, logical flow, and error-free prose.
- **Formatting and Citations (10 points):** Follows guidelines; proper citations.